Arunabha Majumder

+45 71 62 24 00 | Aalborg, Denmark
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder

November 1, 2026

German Aerospace Center (DLR), Institute for the Protection of Maritime Infrastructures, Bremerhaven
Attn: Dr. Maurice Stephan

Application: Teamlead Distributed Cooperative Systems (Vacancy-ID 5677)

Dear Dr. Stephan,

I am applying for the Teamlead Distributed Cooperative Systems position at the DLR Institute for the Protection of Maritime Infrastructures. I am a mechatronics and robotics engineer doing my PhD at Aalborg University, with my thesis expected to be finished in March. What draws me to this role is the chance to take autonomous systems out of the lab and into real field trials at sea.

My strength is designing, building and testing robotic systems that work in the real world. In my PhD I designed a hybrid shoulder exoskeleton, optimised it with a genetic algorithm, integrated the motor, encoder and force sensors, and tested it on a physical setup and with 12 participants. It won the Best Paper Award at IFToMM ISRM 2026. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. To make these experiments work, I implemented an Arduino-based trigger system to synchronise the robot's motion with Qualisys motion capture. Planning complex experiments, building prototypes and getting reliable data from them is the part of research I know best.

I also have experience guiding technical work. Since 2023 I have taught university courses and supervised Bachelor's students in their projects, and I have presented my work at international conferences. I am used to working across disciplines with colleagues in control, electronics and software.

I want to be honest about the gaps. My background is in robot hardware and experiments rather than swarm intelligence and multi-agent decision-making, and my German is not yet at the level you ask for. I am willing to learn German and to grow into the leadership side of the role. If this team lead position is not the right fit for my profile today, I would be glad to be considered for a researcher position in your team, where my experience with prototypes and field-ready hardware could help.

Thank you for considering my application. You can see my work at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder